1 NC code record format accepted by PC-CAM software

NC code reading in the PC-Cam software has been prepared in order to identify a maximum number of postprocessors that are included in CAM software.

NOTE !!!

IT IS NOT POSSIBLE TO PROVIDE COMPATIBILITY OF READING FILES FROM ALL POSTPROCESSORS AND ALL CAD SOFTWARE

NOTE !!!

Performing paths saved in the NC code requires the PC-CAM software in 3D version

Taking into account the development of engineering software and increasing requirements of accuracy and complexity of processing, the main emphasis has been placed on the lack of limits on the length of the code loaded. But when loading G-CODE files into program, limitation, based on maximum available memory for operating system (and task), need to be considered. In case of problems loading the large file NC file should be split into several smaller, or if that does not help, contact the service. The files that contain the machine code should have NCC or NC extension.

2 Characters recognized in the code:

*	character of the beginning of the line with the comment
;	character of the beginning of the line with the comment
_	character of the beginning of the line with the comment
(	the beginning of the comment
)	the end of the comment
G	preparatory function followed by a one or two-digit code
F	feed speed control function
S	rotational speed control function
T	tool changing
X,Y,Z	final coordinates of the straight line or the arc
M	auxiliary function followed by a one or two-digit number
N	the line numbering

3 Initial coordinates

Uwaga:

If the NC code in the first line that contains axes coordinates at the beginning of the file or the first line that contains axes coordinates after tool change (line with T) will not be given the coordinates of all the available axes, the NOT SPECIFIED axes will be used ITS CURRENT position

Example:

T8 – tool exchange

C180 – C axis is moving to 180 degree position other axes stay still

4 Comments

The following signs indicate starting the line with the comment: *, _ , ; (semicolon).

If the interpreter detects one of these characters, it goes automatically to loading the next line of the code. This means ignoring all the characters to the end of the line. To place a comment you can also use characters ( ) - parentheses. In this case the comment is placed inside the parenthesis; the characters outside the parenthesis will be interpreted.

Examples of using comments:

```
 * It is a comment...
 X10 Y20 *It is a comment...
(It is a comment...) X30 Y40
```

5 Feed control.

The F character is followed immediately by a number greater than zero indicating the machine feed speed in mm/min. This speed is compared with a maximum speed of the machine and is entered as the speed of cutting moves of linear and circular interpolation, starting with a given line to cancelling with another F instruction.

6 Speed control - RPM.

The S character is followed by a number greater than zero indicating the rotational speed of the tool. This number is compared with the minimum and maximum rotational speed and saved as the current speed for a given tool (stage). The rotational speed control is possible only by machines with the appropriate communicating module located in the frequency inverter.

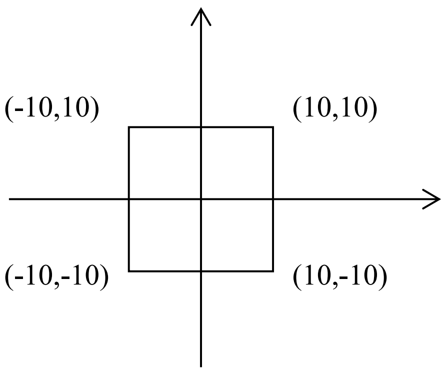
7 Line numbering.

The line numbering is not obligatory. However, you can save the existing tool path by adding the line numbers to it. In this case the lines will be numbered using the N character and the number identifying the next line in the code.

8 Coordinates dimensioning.

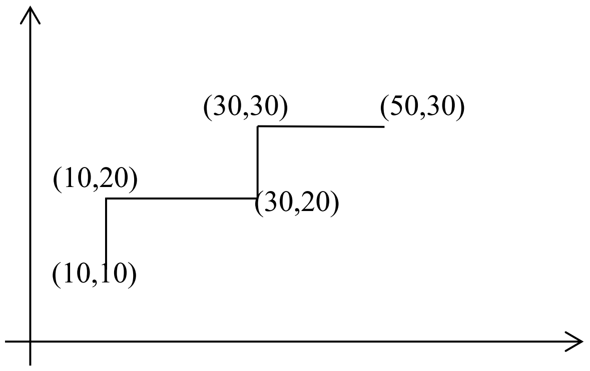
The functions G90 and G91 determine the way of the coordinates dimensioning. In the case of G90 we have absolute coordinates, in the case of G91 the coordinates are incremental. The record in the absolute coordinates means the feed of the machine to a specified point, the record in incremental coordinates - the feed by the value preset. The following examples illustrate two ways of dimensioning.

Absolute dimensioning




```
G90
G0 X10 Y10
G1 X-10
Y-10
X 10
Y10
```

Incremental dimensioning




```
G91
G0 X10 Y10
G1 Y10
X20
Y10
X20
```

The default value which is valid from the start of the program is G90. If the entire code is given in absolute terms, there is no need to call the G90 instruction. When all values are given in incremental coordinates (G91 at the beginning of the code), the default starting point is X = 0 Y = 0 Z = 0

9 Coordinates dimensioning units.

The functions **G70** and **G71** change dimensioning units of coordinates. G71 means dimensioning in millimetres, G70 in inches. The default unit valid from the start of the code are millimetres. If the entire saved code is given in millimetres, there is no obligation to give G71.

10 Approaching motion (G0)

Calling this function causes the motion of the machine to the destination point specified by the XYZ coordinates at a maximum speed of the machine. This speed is set in the Options menu -> Machine Settings separately for the X,Y axes and the Z axis.

Note! The value of the maximum speed should not be increased without consulting the manufacturer. Approaching speed is regulated by the knob on the control panel in the range from zero to 145% without the possibility of exceeding the maximum speed. The G0 instruction is performed until it is cancelled by the G1, G2 or G3 instructions.

11 Linear interpolation (G1, G01)

Calling the G1 function causes the movement of the machine along a straight line to the destination point specified by the coordinates XYZ. This movement takes place with a operating speed specified by the the F parameter, for example:

```
G1 X10 Y20 Z30 F300
```

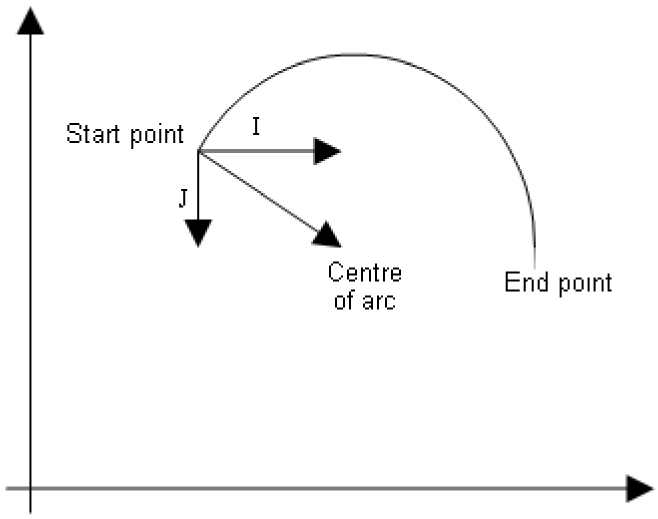
means the movement along a straight line to the point (10,20,30) with the speed of 300 m/min. If in a given line the speed is not specified, the previous value of speed is assumed. It is similar in the case of any value of the feed, for example:

```
G1 X10 Y20 Z30 F300
X50
```

means the movement along a straight line to the value (10,20,30), and then to the value (50,20,30). The operating feed (G1) applies to all lines until it is cancelled by G0, G2 or G3 instructions. The value of the feed speed can be adjusted from 0-100% with the knob on the control panel. Operating speed cannot be greater than the maximum speed of the machine specified in the Options menu -> Machine Settings.

12 Circular interpolation (G2, G3)

The G2 or G02 function means the feed along the arc in the clockwise direction to the destination point. The G3 or G03 function means the counterclockwise movement . You can place the arc on the XY, XZ or YZ planes. There are two ways to write circular interpolation.



Circular interpolation with J,I,K parameter

The I, J, K coordinates are the values of the displacement of the point of the beginning of the arc to the centre of the arc.

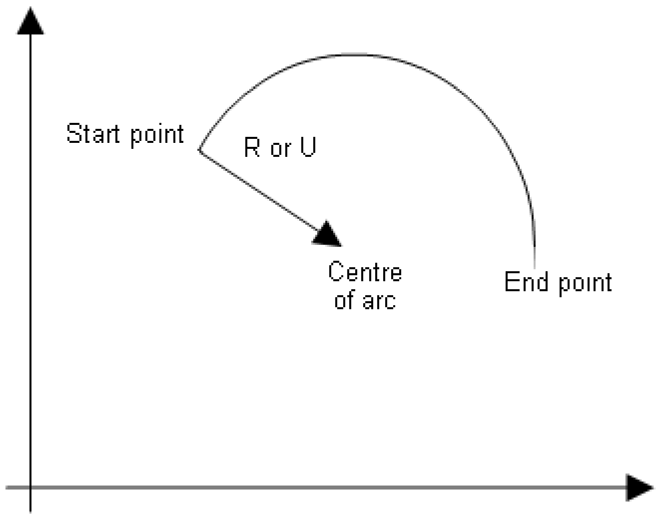
1. I - X component of the displacement
2. J - Y component of the displacement
3. K - Z component of the displacement

If the arc is located on the XY plane, the K component is set to zero automatically. If the end point will coincide with the end point, the full circle will be made.

An example record of circular interpolation:

```
G0 X50 Y50
G2 X50 Y60 I0 J5
```

means approaching the point (50.50), and next the feed along the semi-circle from the start point to the end point. Coordinates of the arc centre are (55.50) (X coordinate of the start point plus I and Y coordinate of the start point plus J). Approaching will take place in the clockwise direction.



Circular interpolation with R (or U) parameter

The G2 instruction sets the clockwise direction, instruction G3 means the feed goes in the counterclockwise direction. The R (or U) parameter means the value of the radius of the arc appointed by the following rules:

• If the length value is preceded by - (minus), the central angle of the arc is greater than 180 degrees.

• If the length value is not preceded by any character, the central angle of the arc is less than or equal to 180 degrees.

13 Feed on the helical path

The feed on the helical path is possible only in the XY plane. To perform this feed we declare simultaneously all the three parameters of the position of the destination point and the parameter defining the movement along the arc G2 or G3. For example:

```
G0 X10 Y0 Z0
G2 X-10 Y0 Z-5 I-10 J0
G3 X10 Y0 Z-10 I10 J0
G2 X-10 Y0 Z-15 I-10 J0
G3 X10 Y0 Z-20 I10 J0
```

means approaching the point (0,0,0), and next plunging on the helical path to the depth of -5 and further successively to the depth of -10, -15 and -20. At each plunging the machining direction is changed.

14 The choice of the machining plane

The preparatory functions G17, G18, G19 are used to select the machining plane.

If you do not declare any value, the G17 value is assumed as default. The plane selection is important in circular interpolation.

G17 - XY-plane of processing,

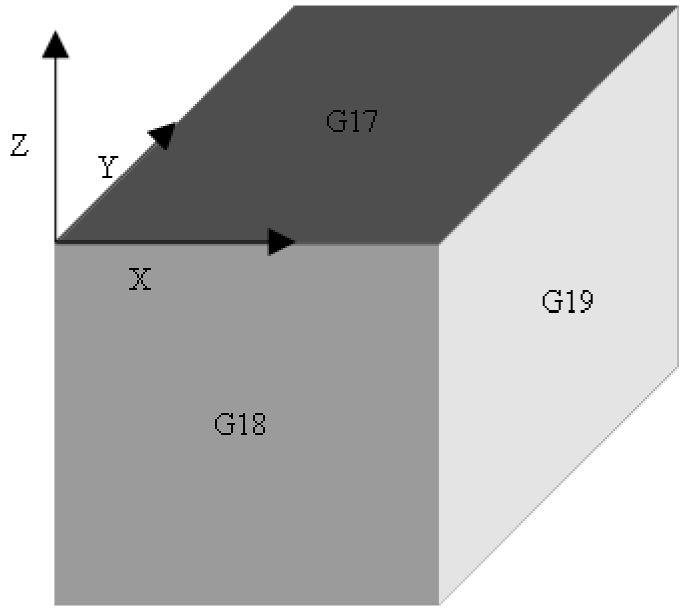
G18 - ZX-plane of processing,

G19 - YZ-plane of processing,

NOTE !!!

In the case of the helical path motion the machining plane is always G17.

The following figure shows the arrangement of planes selected with G17, G18, G19 functions.



Example of machining on the plane other than default one:

```
G0 X0 Y0 Z0
G18
G3 X10 Y0 Z0 I5 K0
G1 Y5
G2 X0 Y5 Z0 I-5 K0
```

The above statement should be understood as follows: approaching the point (0,0,0); switching the machining to the XZ plane; the movement along the arc to the point (10,0,0); the movement along a straight line to (10,5,0) and the return arc (0,5,0). In machining on the XZ plane the displacement parameter of the arc centre along the Y-axis - J has been replaced by the K - displacement parameter of the arc along the Z-axis.